

Module 5: Scientific communication and assessment

Instructional time: 30 min (10 knowledge check, 20 capstone). Bloom: Evaluate, Create. Notebook: none. Ships as: document only.

1. Overview

Modules 1 to 4 produced a set of results: a verified cohort, an integrated object with four compartments, a multi-omic interpretation, and a cross-resource comparison. Module 5 asks what can honestly be claimed from work of that kind, and then asks you to apply the same approach to a biological question of your own.

Key components. Three principles of fidelity that govern how a computational result becomes a scientific claim; a capstone in which you design an analysis for a new question using Common Fund resources; and a ten-item knowledge check.

Learning objectives. After this module you will be able to:

- Apply the principles of data, method and context fidelity to a computational result.
- Design an analysis that carries the approach used in this course to a new biological question.
- Identify which Common Fund resources bear on a question and what each one contributes.

2. Three principles of fidelity

A computational result becomes a scientific claim through three commitments. Each one is visible in the work you have just done.

Fidelity of data: report at the granularity the analysis produced

An analysis produces its result at a particular resolution, and the report should not exceed it.

Module 4 tested whether pathways concordant between the disease and aging contrasts shared enough of their driving genes to be described as one mechanism. Leading-edge clustering merged none of the five. The result is therefore five separate concordant pathways, grouped only by direction: cholesterol biosynthesis and SREBF activation down, EMT, ECM organization and keratinization up.

Describing those five as "sterol synthesis and epithelial remodeling" would be more satisfying to write, and it would assert coordinated regulation that the leading-edge overlap does not support. The same applies in the aging and flight comparison, where clustering merged exactly one pair of terms and left oxidative phosphorylation, fatty-acid beta-oxidation and mitochondrial protein import distinct. All four are mitochondrial, and shared localization is not evidence of shared regulation. "Mitochondrial metabolism is down" is a fair summary sentence; it is not a mechanism claim.

Fidelity of method: ensure reproducibility of research and findings

A result that cannot be regenerated is an anecdote. Reproducibility here is specific and checkable.

Name the exact records. HuBMAP publishes a processed derivative for each primary dataset, and Donor_1 has two published derivatives; this course pins one and says which. Cite the primary DOI, state the release analyzed, and a reader can return to your starting point.

Record the decisions the result depends on. Reference levels in this course are set explicitly, and that is not a formality: alphabetical factor ordering would have inverted three of the four contrasts in Module 4. Quality-control thresholds, gene-set libraries, random seeds and the number of permutations all belong in the record for the same reason.

Ship the artifacts a claim rests on. Every number in these documents is read from a committed table rather than transcribed, so a reader can check any figure against the file that produced it.

Fidelity of context: bound interpretation to evidence and disclose limitations

A claim carries the conditions under which it holds.

Distinguish a negative result from an uninterpretable one. In the flight contrast every pathway that passed threshold moved in the same direction. That unanimity is a warning about the contrast, not a finding about spaceflight. Three explanations were tested and rejected, and the honest report is that the contrast cannot answer a directional question, which is a different statement from finding no effect.

Keep the methods result separate from the biology result. The calibration checks in Module 4 passed convincingly. A sound pipeline pointed at a contrast that cannot answer the question is still sound, and still cannot answer the question.

State the conditions plainly. The HuBMAP age comparison is two donors against two, and is presented as a design illustration rather than a powered result. Disclosing that is what allows a reader to weigh it correctly.